

Neelkumar K. Shah

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PROFESSIONAL SUMMARY

Blockchain security engineer with 4+ years of experience across blockchain development, full-stack engineering, and cybersecurity. Built and hardened Hyperledger Fabric and Ethereum systems for India's National Blockchain Service initiative under the Ministry of Electronics and Information Technology (MeitY). Developed a full-stack security assessment platform (React, Node.js, MongoDB) and an automated database vulnerability scanner. Authored 5 peer-reviewed IEEE publications on blockchain security.

SKILLS

Blockchain: Hyperledger (Fabric, Sawtooth), Ethereum (Solidity, Truffle, Ganache)

Development: Python, Node.js, Express, React, MongoDB

Security: Vulnerability Assessment & Penetration Testing (Web Applications, APIs), Smart Contract Audits (Ethereum, Hyperledger), Static Source Code Analysis

PROFESSIONAL EXPERIENCE

School of Applied Sciences, Engineering and Technology - Rashtriya Raksha University

Assistant Professor | Gandhinagar, Gujarat, India | Jun 2026 - Present

- Delivered a full-day training session on blockchain and cryptocurrency fundamentals.
- Prepared Board of Studies (BoS) documentation for the PG Diploma in Blockchain Technology program.

C-DAC: Centre for Development of Advanced Computing

Project Engineer | Pune, Maharashtra, India | Oct 2021 - Jun 2026

- Spearheaded R&D under the Ministry of Electronics and Information Technology (MeitY) through design and development of a unified blockchain framework to support National Blockchain Service initiatives.
- Architected and configured Hyperledger Fabric networks, managing node orchestration, organizational setup, and client SDK integration.
- Authored and presented 5 peer-reviewed papers at IEEE ICBDS and I2CT conferences on blockchain security.
- Led development of a Blockchain Security Assessment Portal using React.js, Node.js, Express.js, and MongoDB to automate security evaluations for blockchain applications, APIs, containers, and databases.
- Designed and developed NBFDBAudit, a Python-based automated tool that detects 18+ security vulnerabilities in MongoDB and CouchDB databases used within blockchain environments.
- Developed and published API security testing frameworks for Hyperledger Fabric and Sawtooth environments.
- Developed and deployed Hyperledger Fabric blockchain networks including peers, channels, MSP, and certificate authority configuration.
- Built REST APIs for blockchain services integration with enterprise systems.
- Performed Vulnerability Assessment and Penetration Testing (VAPT) on blockchain applications and APIs.

Huptech Web Private Limited

Jr. UI Designer and Developer | Ahmedabad, Gujarat, India | Jun 2019 - Sep 2019

- Developed responsive web interfaces using HTML, CSS, JavaScript, jQuery, and Bootstrap.

ACADEMIC PROJECT

Development and Deployment of E-waste Management Modules Using Blockchain Technology

Graduate School of Engineering and Technology, GTU (M.E. Dissertation) | Jun 2020 - Jul 2021

- Designed and built a decentralized Ethereum DApp using Solidity, Truffle, Ganache, and IPFS.
- Implemented secure, tamper-resistant ownership tracking across the e-waste lifecycle.
- Integrated a React.js frontend with MetaMask wallet connectivity for on-chain interactions.
- Conducted threat modelling and performance optimization of the decentralized architecture.

PUBLICATIONS

A Threat Modeling Approach for Securing Blockchain Framework

2025 4th IEEE ICBDS | ieeexplore.ieee.org/document/11378344 | 17 Feb 2026

Security Assessment Framework and Evaluation for Blockchain Applications (SAFE-Block)

2024 3rd IEEE ICBDS | <https://ieeexplore.ieee.org/document/10837302> | 17 Jan 2025

Efficient Solution for NoSQL Database Security in Blockchain-Based Applications

2023 2nd IEEE ICBDS | <https://ieeexplore.ieee.org/document/10346312> | 15 Dec 2023

Smart Contract Vulnerability Detection Techniques for Hyperledger Fabric

2023 8th IEEE I2CT | <https://ieeexplore.ieee.org/document/10126362> | 23 May 2023

Smart Contract - Security Assessment Integrated Framework (SC-SIF) for Hyperledger Fabric

2022 7th IEEE I2CT | <https://ieeexplore.ieee.org/document/9824439> | 18 Jul 2022

EDUCATION

Master's Degree in Cybersecurity

Graduate School of Engineering and Technology, GTU, Ahmedabad, Gujarat, India | Sep 2019 - Jul 2021

Bachelor's Degree in Computer Engineering

Marwadi Education Foundation, Rajkot, Gujarat, India | Aug 2015 - May 2019

CERTIFICATION & ACHIEVEMENTS

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- Certified Blockchain Security Professional - Blockchain Council, USA
 - Certificate of Completion for Certified DevSecOps Engineer Training - EC-Council, USA
 - Recognized as Performer of the Month - C-DAC, Pune, Maharashtra, India
 - Blockchain Technology for Government Officials - Future Skills PRIME, NIELIT, Kolkata, India